

Vijay Ramakrishnan

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EDUCATION

Georgia Institute of Technology([Georgia Tech](#))
Master of Science in Computer Science - CGPA : **4.0/4.0**

Atlanta, Georgia
Aug 2025 - Present

Indian Institute of technology Palakkad ([IIT Palakkad](#))
Bachelor of technology in Computer science Engineering

Kerala, India
Aug 2021 - July 2025

SKILLS

Programming Languages: Python, C++, C, Java, SQL, MATLAB

ML: PyTorch, TensorFlow, Scikit-learn, OpenCV, LangChain

Systems & Infrastructure: Django, Docker, Celery, Redis, gRPC, ROS2, Git, Linux

Databases: PostgreSQL, MongoDB, Qdrant

EXPERIENCE

AI Engineer Intern - MagicLemp:

Paris, France *Dec'25 - June'26*

- **Engineered a SharePoint-to-Qdrant synchronization pipeline** using Microsoft Graph API, Celery, and Redis, implementing incremental freshness checks, content-hash deduplication, distributed API concurrency control, and rate limiting to avoid redundant indexing and embedding workloads.
- **Built RAG and web-search agents** as streaming gRPC/Protobuf services, supporting permission-aware retrieval, query reformulation, source-grounded citations, and ephemeral Qdrant collections; contributed to a production platform spanning **13 document formats** and multiple Dockerized services.
- **Optimized permission-aware vector retrieval** in Qdrant by indexing high-cardinality file identifiers, reducing query latency from **25–52s to 1.6 to 1.8s** on over **2 million vectors** with filters spanning up to **1 million file IDs**.

Software Engineer Intern - Tata Research Group:

Bangalore, India *June'24 - Aug'24*

- Engineered **PEOA** in Python as a modular LLM orchestration framework, implementing task decomposition and dynamic routing across specialized language models and computational tools; integrated **Graph-RAG** for domain-grounded retrieval and reasoning.
- Built **PatExpert**, a multi-agent patent analysis system with a **Meta-Llama-3.1-405B** orchestrator and specialized **GPT-4o-mini** agents for classification, summarization, claim generation, and multi-patent analysis; implemented LLM-as-a-Judge feedback loops for automated response refinement.
- Built evaluation pipelines for **LLM-based automation systems**, benchmarking PEOA against domain-specific baselines and achieving **73.68% / 74.13% Exact Match** on MathComp / ChemProc; work contributed to papers submitted to **AAAI and NeurIPS**.

PROJECTS

Autonomous Treasure Search, Multi-Sensor Mobile Manipulator

Aug'25 - Dec'25

- Engineered a **ROS2/CoppeliaSim** perception and control system integrating multi-sensor treasure detection, an Ackermann-steered vehicle, and a 4-DOF manipulator for autonomous target localization and retrieval.
- Trained an **MLP** for metal-detector distance estimation and **CNN-based imitation-learning controllers** for vehicle steering and arm positioning, achieving **0.59 MSE** in target localization.

Humanoid Locomotion using Reinforcement Learning

Aug'24 - Dec'24

- Developed a bipedal locomotion controller in **MuJoCo Humanoid-v4** using **Proximal Policy Optimization (PPO)**, training a policy to coordinate balance and locomotion in a high-dimensional continuous-control environment.
- Evaluated training dynamics using **reward, entropy, and value-loss metrics**, diagnosing policy convergence and stability to achieve a stable walking gait.

LEADERSHIP

IAC Coordinator: Led a volunteer team at Petrichor'24 to host Mr. R. Hutton, head of ISRO's Gaganyaan mission, as a guest speaker during IIT Palakkad's annual techno-cultural fest, and coordinated the Industrial Academia Conclave (IAC 5.0) connecting students with industry professionals.

CognifAI Club Mentor : Mentored freshmen in the Machine Learning club of IIT Palakad - CognifAI, introducing them to early concepts of Deep Learning and guiding their participation in institute-level Hackathons and competitions.